

Advanced programming Assignment

ROLL NO: CSB24065

NAME: Gourisankar Bhuyan

Session: Spring

Year:2026

Question:

Build a user onboarding validation module for a platform. Your objective is to create a core validation class that processes incoming application data—specifically a user's email address and age—and enforces strict business constraints before allowing a registration to complete. The system must ensure that the email string is neither null nor empty, and that it conforms to a standard email format matching the regular expression patterns outlined in the slides (containing a valid identifier, an @ symbol, and a domain name). Additionally, the system must enforce a strict age restriction where applicants must be at least 18 years old to create an account.

Implementation Rules

If choosing Java: You must implement a checked exception named `InvalidEmailException` and an unchecked (`RuntimeException`) exception named `UnderageException`. Create a `RegistrationService` class containing a method `public boolean registerUser(String email, int age) throws InvalidEmailException`. You must include an internal assert statement to guarantee that the inputs are not processed if the system context is invalid. Finally, write a JUnit 5 test suite named `RegistrationServiceTest` that uses a `@BeforeEach` setup method, validates successful registrations, and uses `assertThrows` to verify that both custom exceptions are thrown under incorrect conditions.

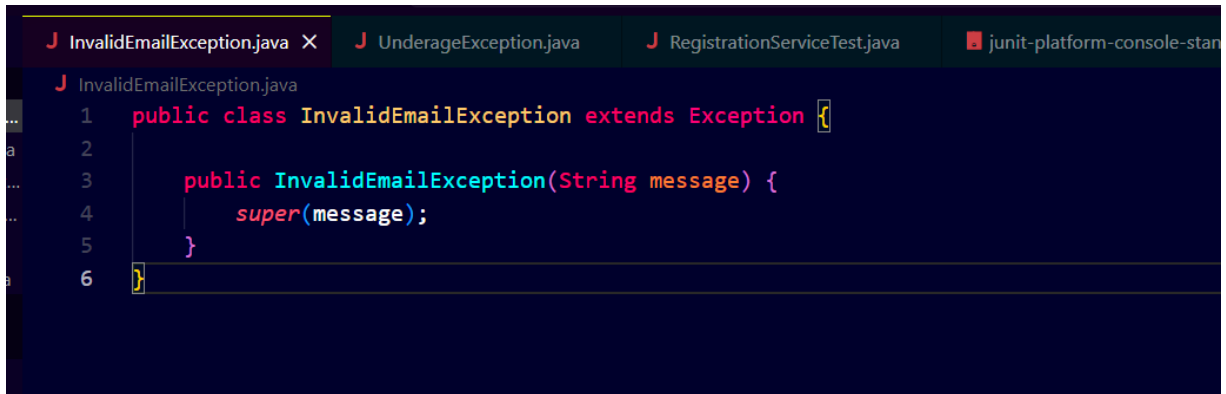
If choosing Python: You must implement a custom exception named `InvalidEmailError` and another named `UnderageError`, both inheriting from the appropriate built-in exception classes. Create a `RegistrationService` class containing a method `def register_user(self, email: str, age: int) -> bool`. Use an internal assert statement to verify basic state invariants. Finally, write a pytest suite using a shared `@pytest.fixture` for configuration, validating successful workflows, and utilizing `pytest.raises` to assert that your custom errors are raised appropriately during invalid inputs. You must have the followings:

Custom Exception Design: Correctly establishing checked vs. unchecked hierarchies (Java) or appropriate base class inheritance (Python) with descriptive, dynamic error messages.

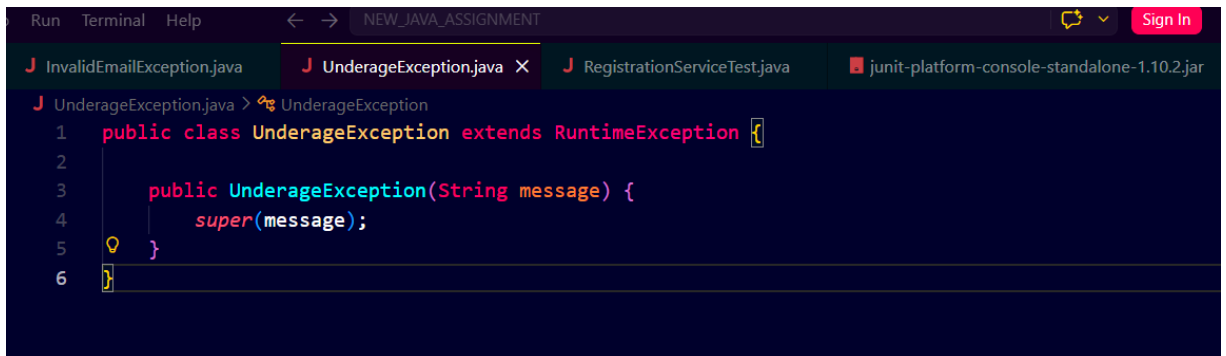
Core Service Validation: Implementing the regex parsing, age boundary checks,

invariant assertions, and proper exception triggering. Unit Testing Suite: Writing comprehensive test cases using the correct framework assertions, proper test lifecycle setup (fixtures/before-each), and targeted exception testing.

Answer:



```
InvalidEmailException.java X UnderageException.java RegistrationServiceTest.java junit-platform-console-standalone-1.10.2.jar
InvalidEmailException.java
1 public class InvalidEmailException extends Exception {
2
3     public InvalidEmailException(String message) {
4         super(message);
5     }
6 }
```



```
Run Terminal Help NEW_JAVA_ASSIGNMENT Sign In
InvalidEmailException.java UnderageException.java X RegistrationServiceTest.java junit-platform-console-standalone-1.10.2.jar
UnderageException.java > UnderageException
1 public class UnderageException extends RuntimeException {
2
3     public UnderageException(String message) {
4         super(message);
5     }
6 }
```

```
Run Terminal Help < -> NEW_JAVA_ASSIGNMENT Sign In
J InvalidEmailException.java J UnderageException.java J RegistrationServiceTest.java X junit-platform-console-standalone-1.10.2.jar settings.json J RegistrationSen
J RegistrationServiceTest.java > % RegistrationServiceTest > testSuccessfulRegistration()
1 import org.junit.jupiter.api.BeforeEach;
2 import org.junit.jupiter.api.Test;
3
4 import static org.junit.jupiter.api.Assertions.*;
5
6 public class RegistrationServiceTest {
7
8     private RegistrationService service;
9
10    @BeforeEach
11    void setUp() {
12        service = new RegistrationService();
13    }
14
15    @Test
16    void testSuccessfulRegistration() throws InvalidEmailException {
17        boolean result =
18        service.registerUser(email: "user@9999999999999999.com", age: 22);
19
20        assertTrue(result);
21    }
22
23    @Test
24    void testNullEmailThrowsException() {
25
26        InvalidEmailException exception =
27            assertThrows(
28                InvalidEmailException.class,
29                () -> service.registerUser(email: null, age: 25)
30            );
31
32        assertEquals(

```

```
RegistrationServiceTest.java > % RegistrationServiceTest > testNullEmailThrowsException()
6 public class RegistrationServiceTest {
24 void testNullEmailThrowsException() {
32     assertEquals(
33         "Email cannot be null or empty",
34         exception.getMessage()
35     );
36
37
38    @Test
39    void testInvalidEmailThrowsException() {
40
41        InvalidEmailException exception =
42            assertThrows(
43                InvalidEmailException.class,
44                () -> service.registerUser(email: "wrong-email", age: 25)
45            );
46
47        assertTrue(
48            exception.getMessage().contains(s: "Invalid email format")
49        );
50    }
51
52    @Test
53    void testUnderageThrowsException() {
54
55        UnderageException exception =
56            assertThrows(
57                UnderageException.class,
58                () -> service.registerUser(
59                    email: "user@example.com",
60                    age: 16
61                )

```

```
        exception.getMessage().contains(s: "Invalid email form  
    );  
}  
  
@Test  
void testUnderageThrowsException() {  
  
    UnderageException exception =  
        assertThrows(  
            UnderageException.class,  
            () -> service.registerUser(  
                email: "user@example.com",  
                age: 16  
            )  
        );  
  
    assertTrue(  
        exception.getMessage().contains(  
            s: "at least 18 years old"  
        )  
    );  
}
```

OUTPUT:

```
PS C:\Users\gouri\OneDrive\Desktop\NEW_JAVA_ASSIGNMENT> java -jar lib\junit-platform-console-standalone-1.10.2.jar --class-path
>>
>>
Test run finished after 110 ms
[      4 containers found      ]
[      0 containers skipped    ]
[      4 containers started    ]
[      0 containers aborted    ]
[      4 containers successful ]
[      0 containers failed     ]
[      4 tests found          ]
[      0 tests skipped         ]
[      4 tests started         ]
[      0 tests aborted         ]
[      4 tests successful      ]
[      0 tests failed          ]

WARNING: Delegated to the 'execute' command.
         This behaviour has been deprecated and will be removed in a future release.
         Please use the 'execute' command directly.
PS C:\Users\gouri\OneDrive\Desktop\NEW_JAVA_ASSIGNMENT>
```